

FPGA-Vendor Independent AXI BFM



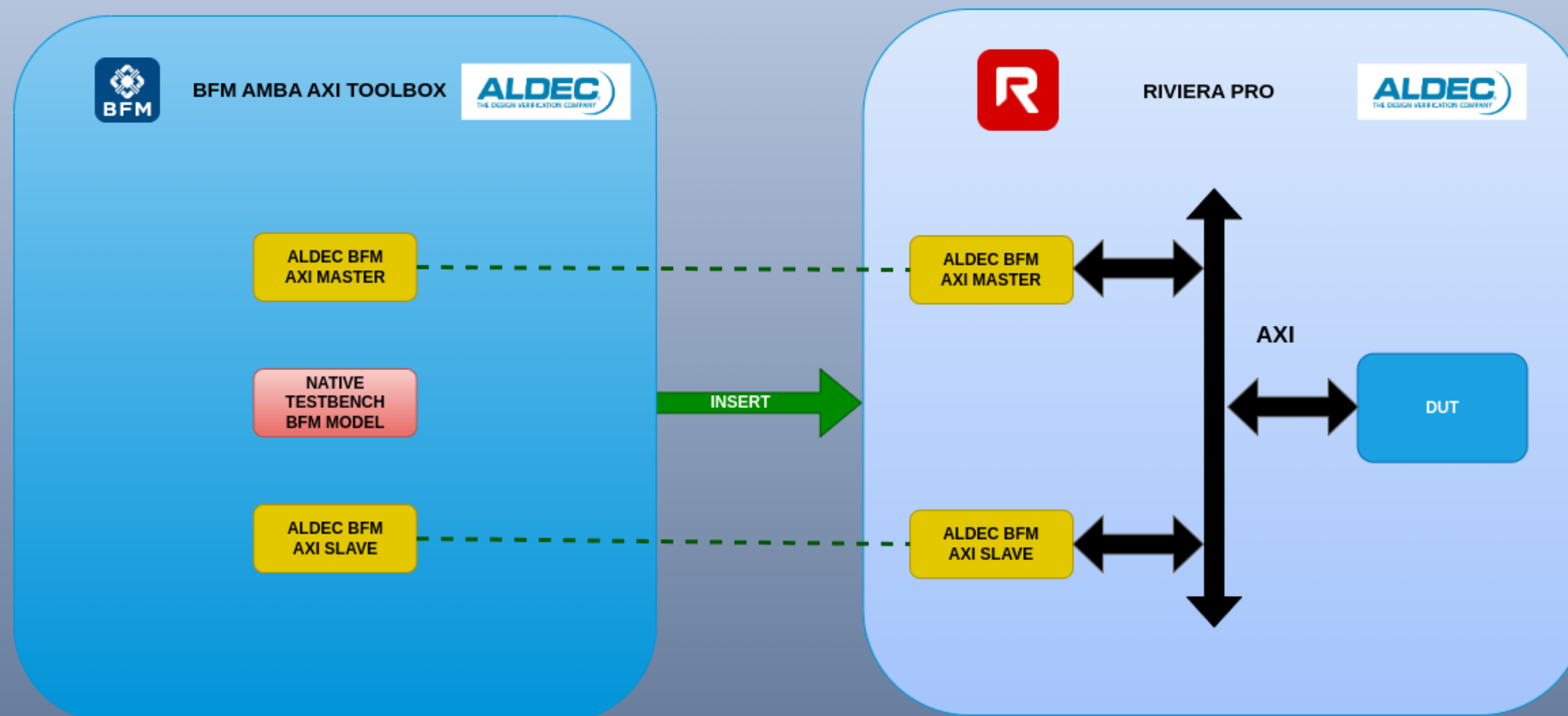
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Concept: how use the Aldec BFM AMBA AXI Toolbox

Native Mode

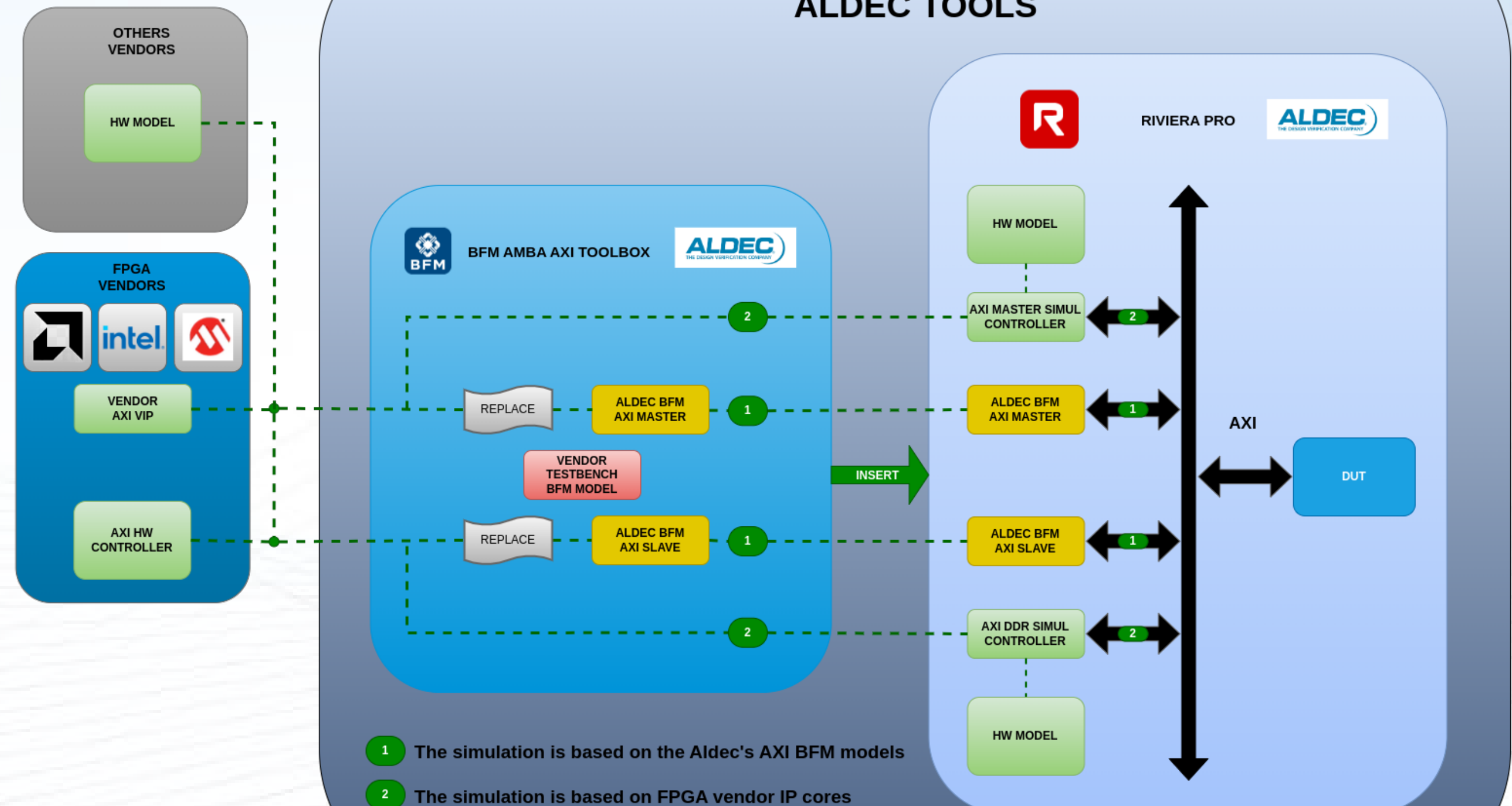
ALDEC TOOLS



The simulation environment is fully based on the Aldec's AXI BFM models

Integration with FPGA Vendor IPs and AXI VIPs

ALDEC TOOLS



- 1 The simulation is based on the Aldec's AXI BFM models
- 2 The simulation is based on FPGA vendor IP cores

The Aldec AXI BFM standard usage

Aldec AXI BFM

1. Task-based Verilog/SystemVerilog
VHDL / Python / HVL / SystemC API

2. BFM protected core

3. AXI Interface

Simulator

Top of the design

Testbench

AXI BFM

Clock and
reset
generator

DUT

Software

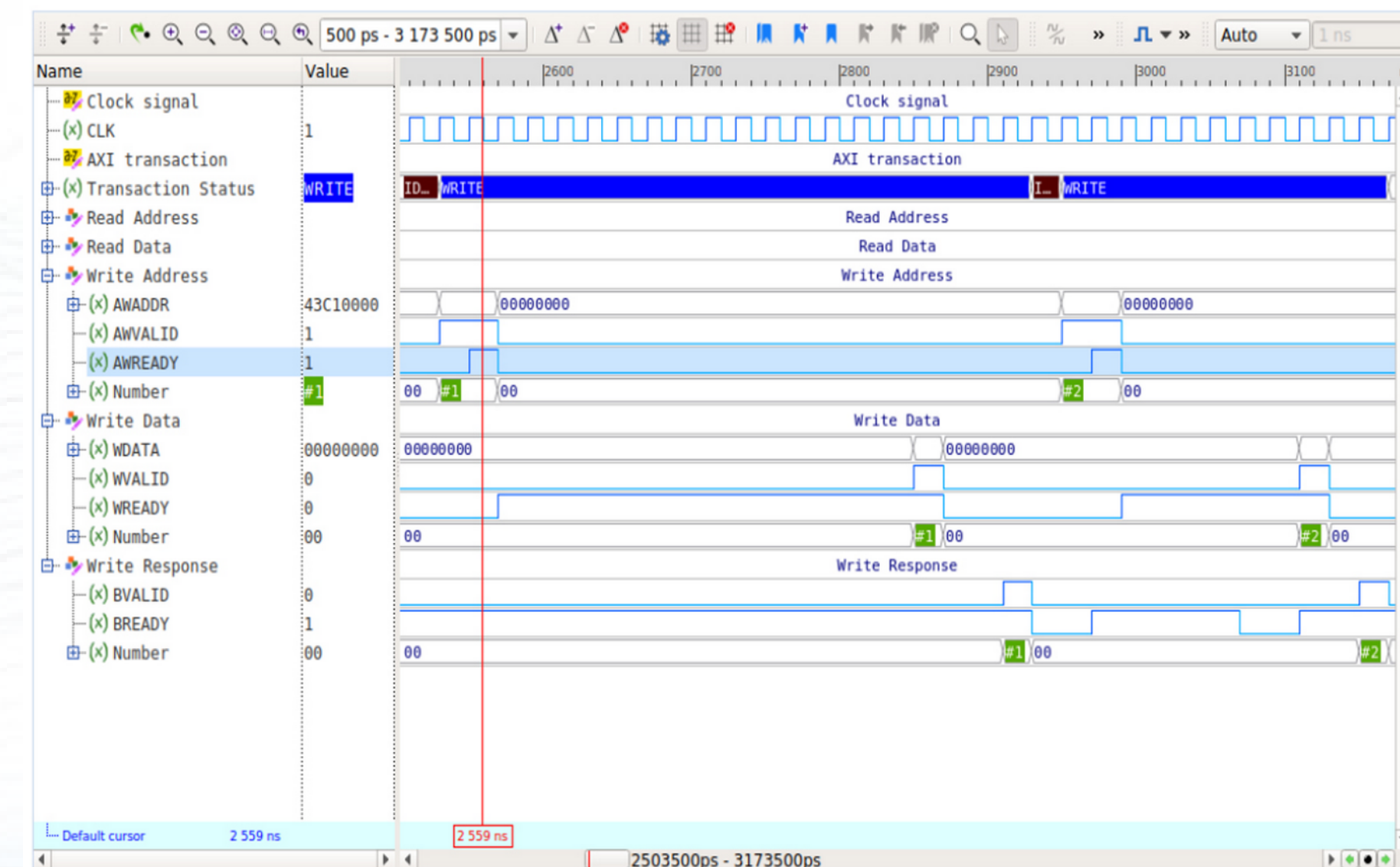
Hardware

AXI BFM
task-based API

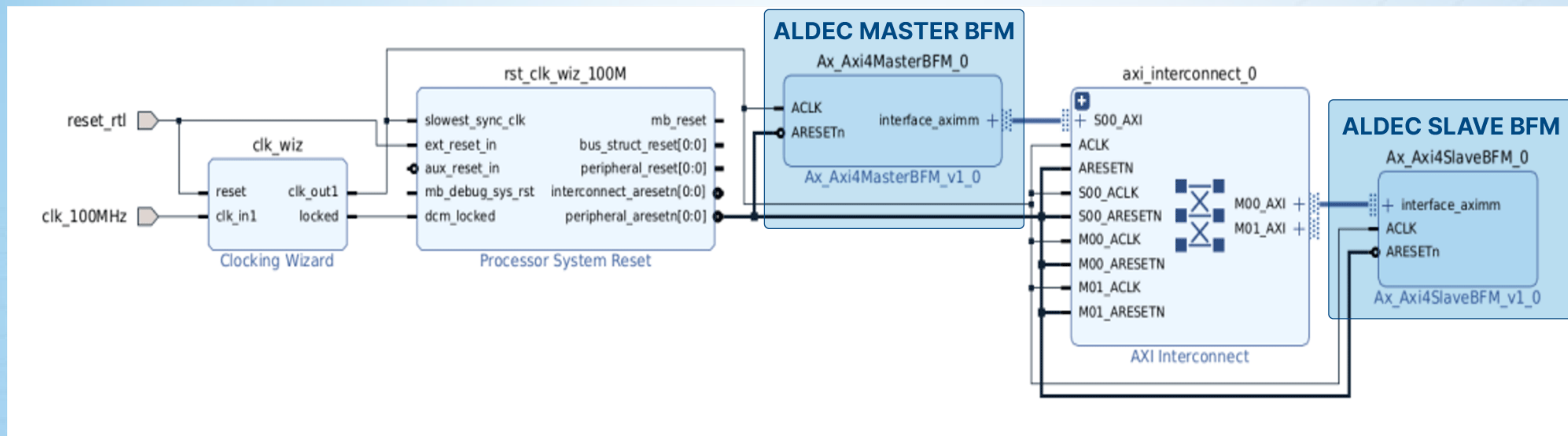
tasks

Testbench

DUT



Supported Vendors



Aldec BFM in Vivado



Supported Vendors

The screenshot displays the Quartus Prime design environment with the following components:

- IP Catalog:** Shows the project hierarchy with 'Basic Functions' and 'Systems' folders. The 'Library' section lists various components, including 'ALDEC AXI4 Master BFM' and 'ALDEC AXI4 Slave BFM'.
- System View:** Displays the 'demo_BFM' system with a 'Path: clock_in'. It shows a table of components and their connections:

Use	Connections	Name	Description	Export	Clock
☒		clock_in	Clock Bridge Intel FPGA IP	Double-click to export	clock_so...
☒		reset_in	Reset Bridge Intel FPGA IP	Double-click to export	clock_in_...
☒		clock_source_0	Clock Source BFM Intel FPGA IP	Double-click to export	clock_sour...
☒		reset_source_0	Reset Source BFM Intel FPGA IP	Double-click to export	clk
☒		Ax_Axi4MasterBFM_0	ALDEC AXI4 Master BFM (New BFM)	Double-click to export	clock_in_...
☒		Ax_Axi4SlaveBFM_0	ALDEC AXI4 Slave BFM (New BFM)	Double-click to export	clock_in_...

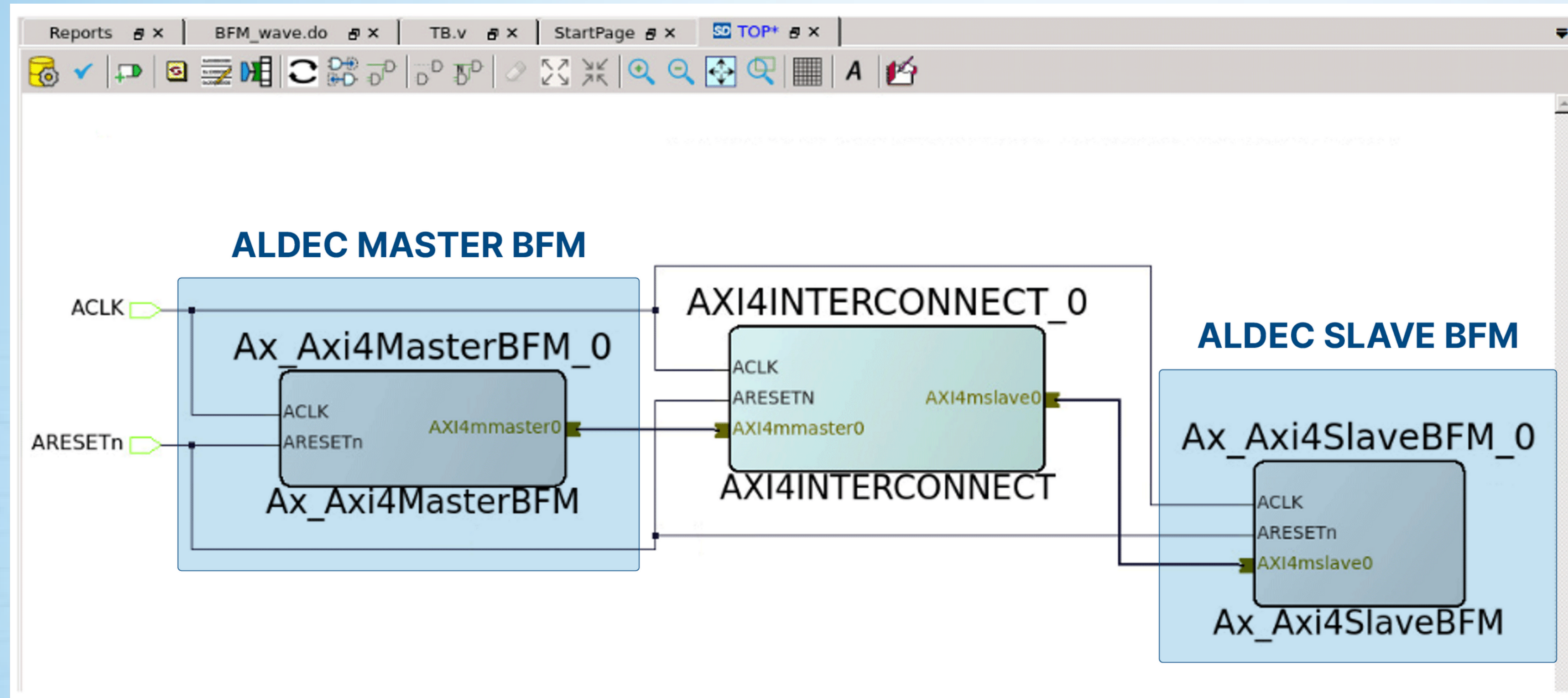
ALDEC MASTER BFM

ALDEC SLAVE BFM

- Component Properties:** Shows the 'clock_in' component with 'Project Settings' and 'Interfaces' tabs. The 'Interfaces' tab lists 'in_clk' and 'out_clk' with 'clock_rate' parameters.
- Interface: in_clk:** Shows the 'Parameter: clock_rate' with a table comparing 'Component Instantiation Value' (50000000) and 'IP File Value' (50000000).
- Presets:** Shows the 'Presets for clock_in' section with a checkbox for 'Show the Presets for Selected Board' and a 'New...' button.


Aldec BFM in Quartus Prime Design

Supported Vendors



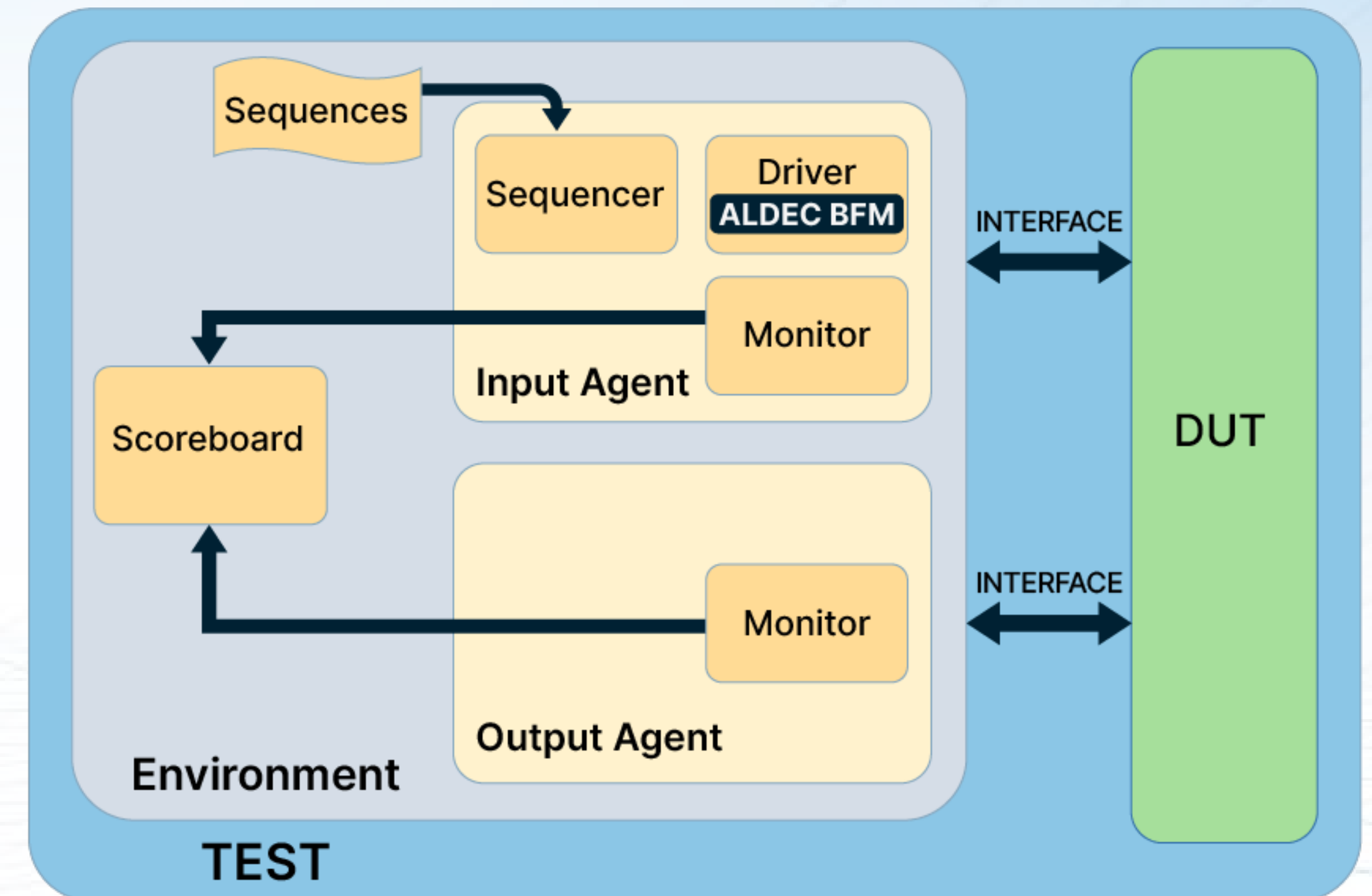
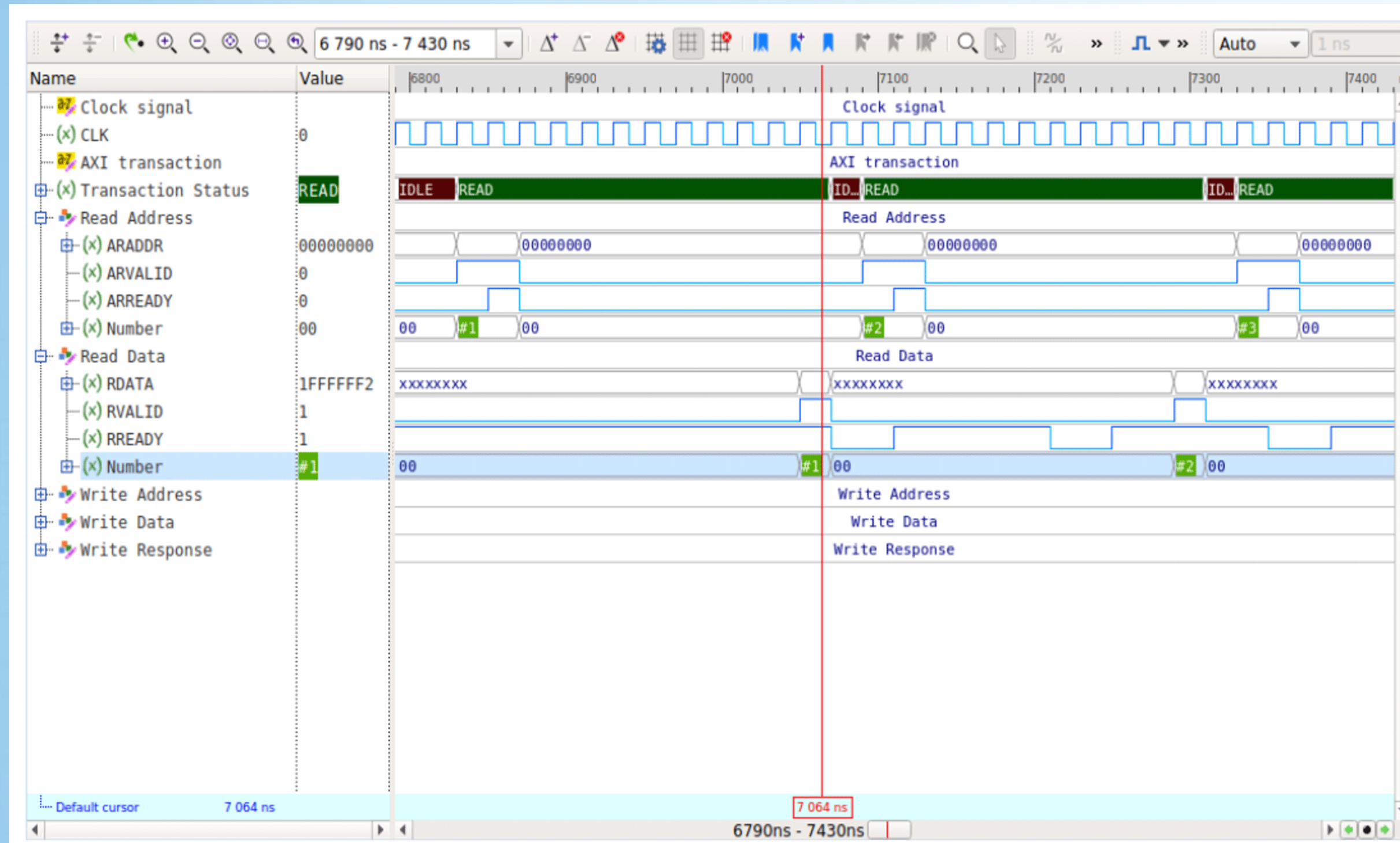
MICROCHIP



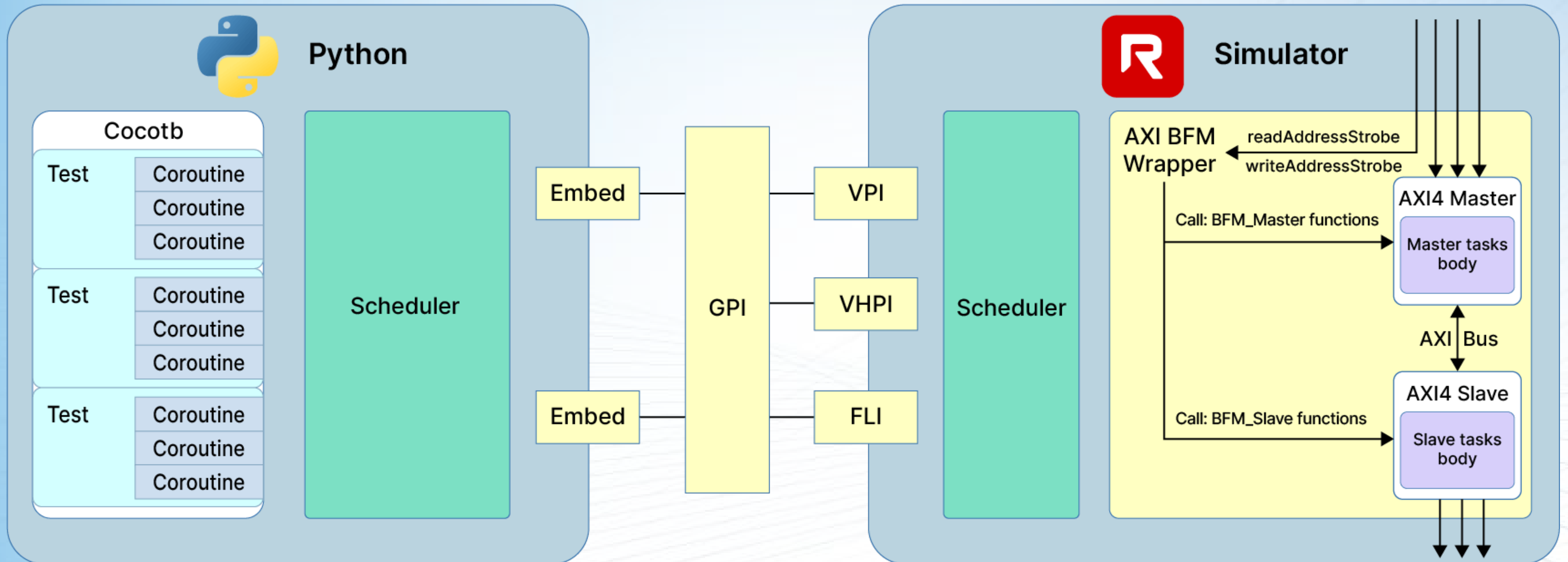
Liberio
System-on-Chip

Aldec BFM in Libero_SoC

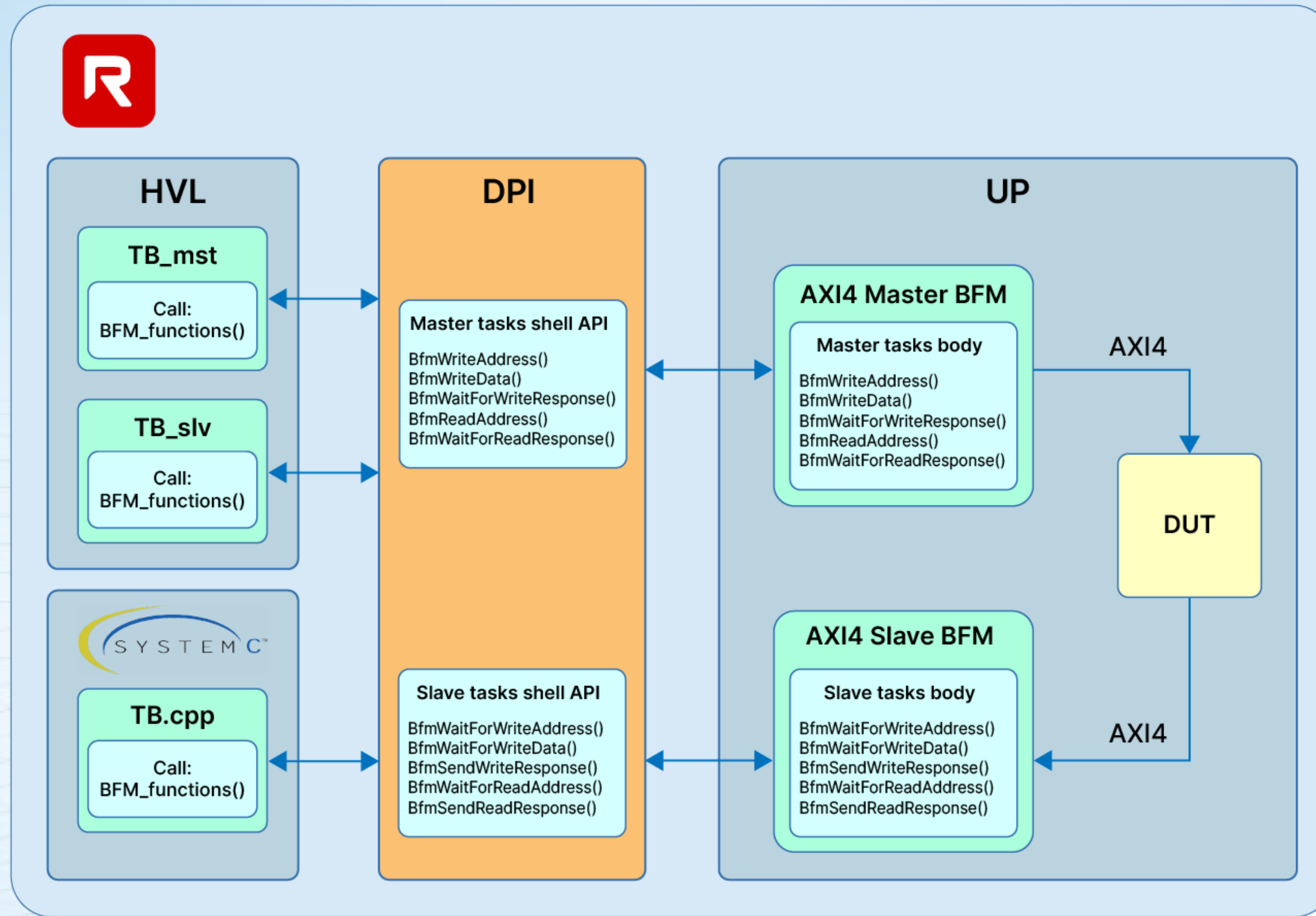
BFM as UVM_Driver



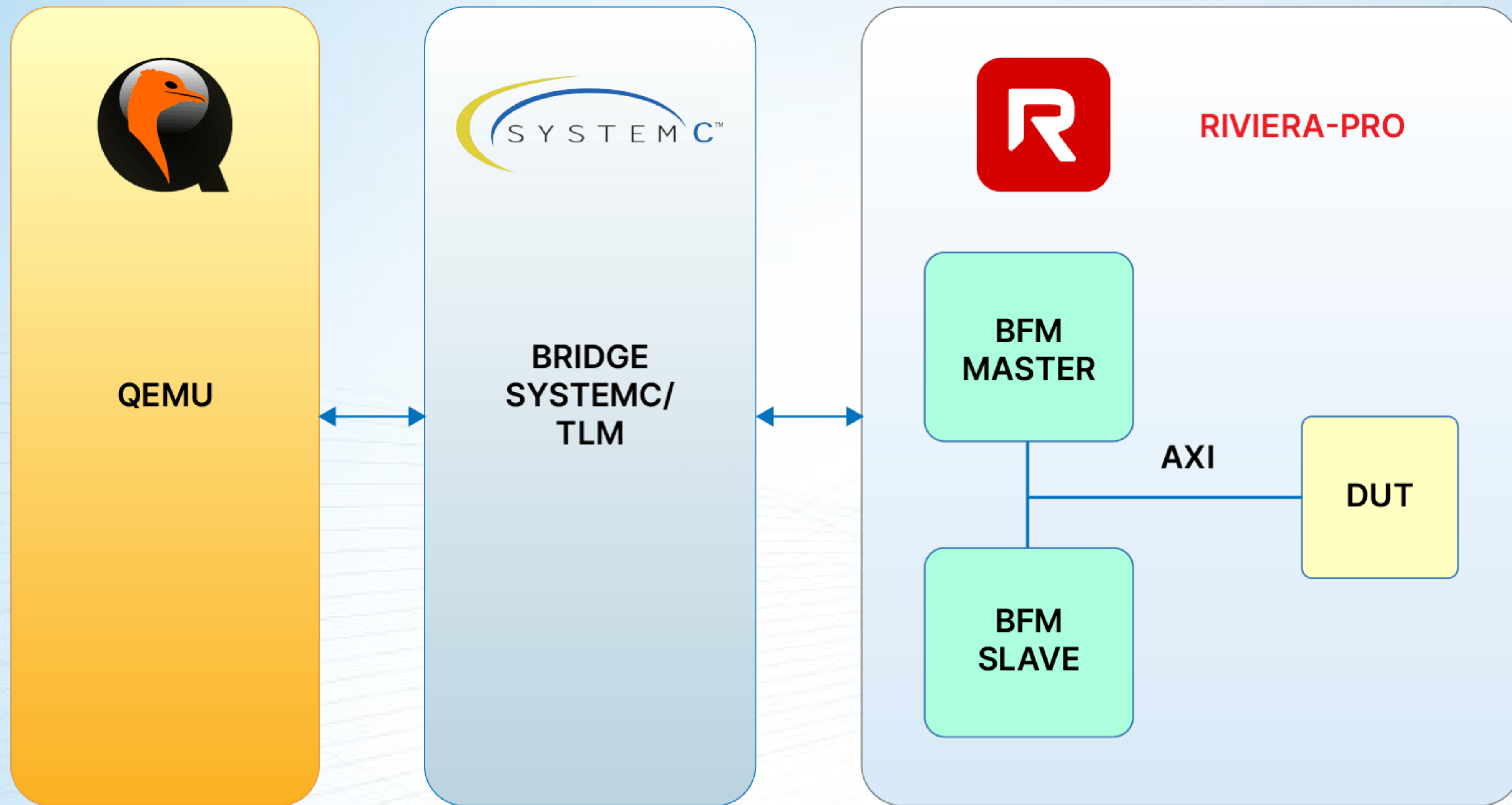
Cocotb with BFM in AXI simulation



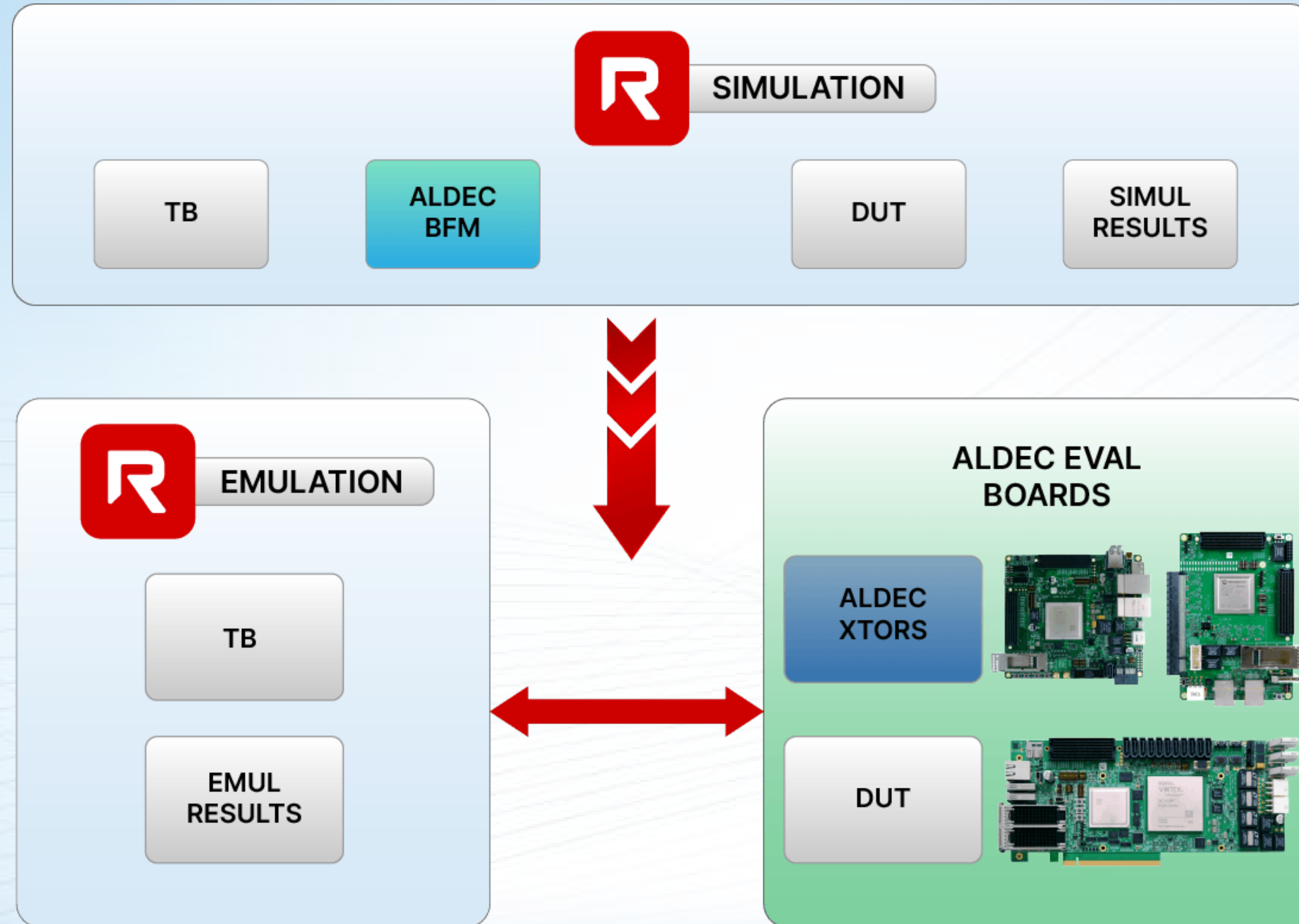
Using BFM through DPI in simulation



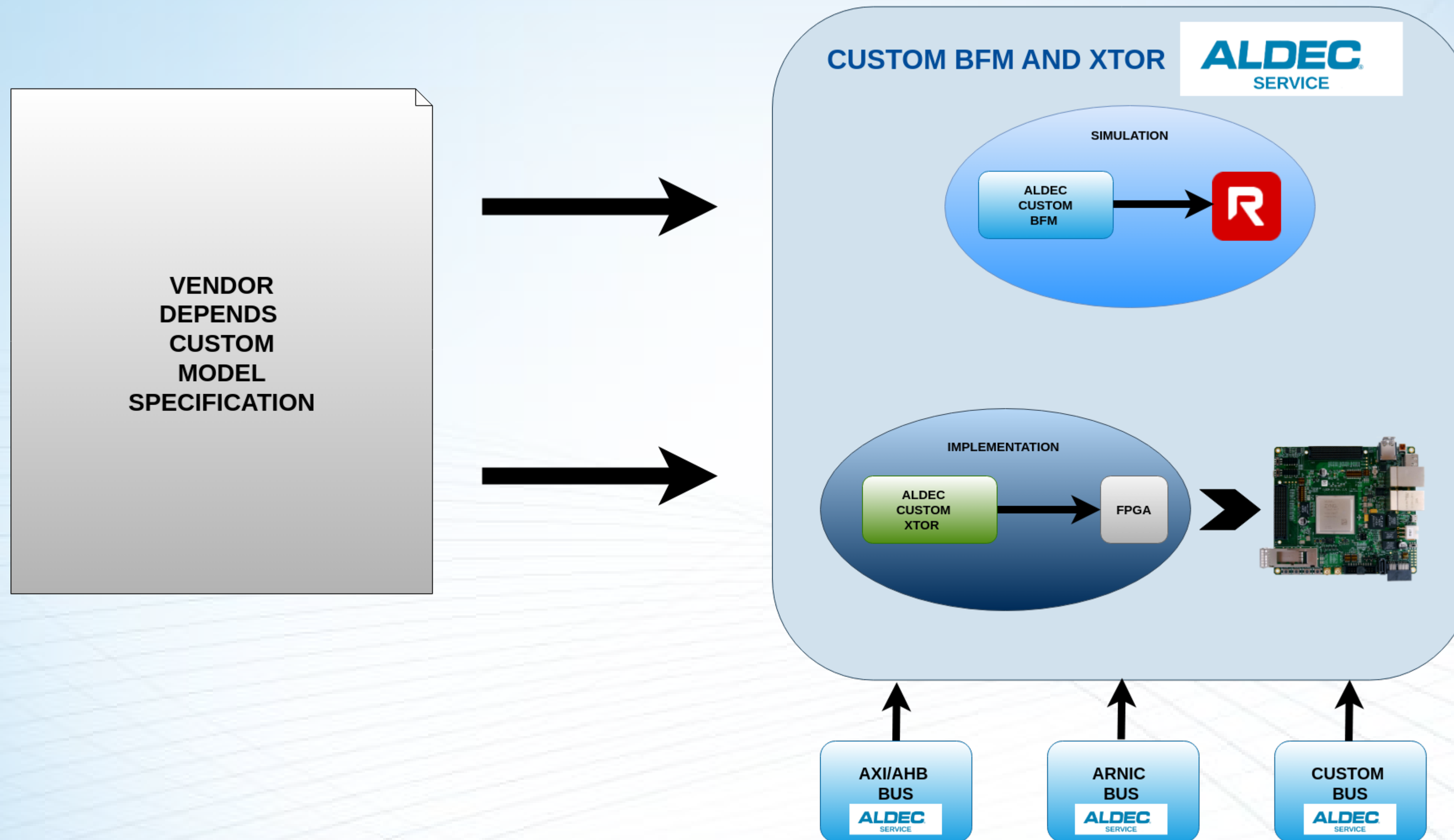
Using BFM with Qemu



HW verification base on XTOR Emulation



Custom BFM and XTOR Service





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